



CAPSTONE PROJECT REGISTER

Class: _____ Duration time: from/.../.... To/... /.....

(*) Profession: < Artificial Intelligence > Specialty: CV ☐ NLP ☒ DS ☐ Other ☐

(*) Kinds of person make registers: Lecturer ☒ Students

1. Register information for supervisor (if have)

	Full name	Phone	E-Mail	Title
Supervisor 1	Đặng Ngọc Minh Đức		ducdnm2@fe.edu.vn	Assoc.Prof.

2. Register information for students (if have)

	Full name	Student code	Phone	E-mail	Role in Group
Student 1					Leader
Student 2					Member
Student 3					Member

3. Register content of the Capstone Project

(*) 3.1. Capstone Project name:

English: Graph-based Multimodal Emotion Recognition in Conversations

Vietnamese: Nhận dạng cảm xúc đa phương thức trong hội thoại dựa trên đồ thị

Abbreviation: GMERC

a. **Context:**

Emotion Recognition in Conversations (ERC) aims to automatically identify the emotional state of each speaker throughout a conversation by jointly analyzing contextual information and interactions among participants. Compared with conventional emotion recognition from isolated utterances, ERC is considerably more challenging because emotions evolve dynamically according to conversational context and speaker interactions. Emotion-aware intelligent systems have become an essential component of many AI applications, including intelligent virtual assistants, customer-service automation, online education, healthcare, and human-computer interaction.

Most existing emotion recognition systems rely on a single modality, such as text or speech, which often leads to incomplete emotional understanding. Recent advances in multimodal learning demonstrate that integrating textual, acoustic, and visual information significantly improves recognition performance. Meanwhile, conversational context and speaker relationships play a critical role in emotion understanding, and Graph Neural Networks (GNNs) have shown remarkable capability in modeling complex relationships among utterances and speakers by representing conversations as graphs — enabling contextual reasoning and capturing long-range dependencies that sequential architectures alone cannot adequately model.

Research Gap:

Although recent graph-based multimodal ERC models have achieved promising performance, they often suffer from one or more limitations, including fixed graph structures, inefficient multimodal fusion, limited contextual reasoning, or high computational complexity. Therefore, investigating lightweight yet effective graph-based multimodal architectures remains an open research direction. This capstone positions students to study this gap and to make a concrete, measurable technical contribution rather than merely re-implementing an existing system.

b. Objectives:

The objectives of this capstone project are:

- Study the fundamentals of Emotion Recognition in Conversations (ERC).
- Investigate recent Graph Neural Network architectures for conversational modeling.
- Explore multimodal feature extraction from textual, acoustic, and visual modalities.
- Construct conversation graphs that capture temporal and speaker relationships.
- Develop a graph-based multimodal emotion recognition model.
- Compare different multimodal fusion strategies and graph architectures.
- Evaluate the model on benchmark datasets using standard evaluation metrics.
- Develop a prototype application for conversation emotion analysis.

Research objective (contribution):

Propose and experimentally validate at least one lightweight improvement over an existing graph-based multimodal ERC baseline, demonstrating a measurable gain (or an equivalent efficiency benefit) supported by ablation and error analysis.

c. Proposed solution:

Data Preprocessing:

Study and preprocess benchmark datasets (MELD as the primary dataset; IEMOCAP optional). Perform data cleaning, conversation organization (grouping utterances by dialogue and speaker), multimodal feature extraction, and a fixed train/validation/test split to ensure consistent, high-quality inputs across modalities.

Feature Extraction:

Use pretrained encoders as frozen feature extractors (without retraining them):

- RoBERTa / BERT for textual features.
- Wav2Vec2 or HuBERT for acoustic features.
- Vision Transformer (ViT) or CLIP for visual features.

The extracted modality features are aligned at the utterance level to form the node representations of the conversation graph.

Conversation Graph Construction:

Represent each conversation as a graph in which utterances are nodes and contextual relationships are edges. Explore several graph construction strategies:

- Temporal connections between consecutive utterances.
- Speaker-aware connections capturing intra- and inter-speaker dependencies.
- Fully connected conversation graphs.

Graph-based Emotion Recognition Model:

Baseline requirement. Each group must reproduce exactly one of the following baselines before implementing any improvement, so that the examination board has a clear reference point:

- DialogueGCN.
- MMGCN.
- GraphCFC.

Investigate GNN architectures such as Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) to learn contextual representations over the conversation graph.

Required improvement (novelty). After the baseline is reproduced, students must implement at least one of the following improvements (or an equivalent technical contribution):

- Adaptive graph construction.
- Adaptive edge weighting.
- Multimodal attention fusion.
- Speaker-aware graph modeling.

- Temporal graph refinement.
- Lightweight graph pruning.
- Graph augmentation.

The improvement must be motivated by the Research Gap above and evaluated against the reproduced baseline under identical conditions.

Multimodal Feature Fusion:

Compare different fusion strategies (early fusion, late fusion, and graph-based cross-modal fusion) to effectively integrate textual, acoustic, and visual representations for robust emotion prediction.

Model Evaluation:

Conduct comprehensive experiments on benchmark datasets, comparing the proposed approach with the reproduced baseline and at least one additional recent deep-learning ERC model. Primary metrics: Accuracy, Weighted F1-score, Macro F1-score, and Confusion Matrix.

Ablation study (required):

To justify that each design choice contributes to performance, students must run ablations such as:

- Without the graph module (sequential model only).
- Without the audio modality.
- Without the visual modality.
- GCN vs. GAT.
- Early fusion vs. late fusion.

Error analysis (required):

The experimental analysis must go beyond aggregate scores and include:

- Error analysis across emotion classes.
- Failure-case visualization from real conversations.
- A discussion of class imbalance and its effect on minority emotions.

Evaluation protocol (for consistency across groups):

Every experiment must report the exact train/validation/test split, random seed(s), hardware configuration, and training and inference time.

Prototype Development:

Build a prototype system capable of loading conversation data and predicting emotions for each utterance, with an enhanced visualization layer:

- Conversation timeline of utterances and predicted emotions.

- Speaker-interaction graph.
- Emotion-transition view across the dialogue.
- Per-prediction confidence score.
- Attention / edge-weight heatmap for interpretability.

d. **Algorithm requirement**

Training

- Datasets: MELD (primary); IEMOCAP (optional).
- Feature extractors: RoBERTa/BERT (text), Wav2Vec2/HuBERT (audio), and ViT/CLIP (visual), used as frozen pretrained encoders.
- Model: conversation graph + GNN (GCN/GAT) with multimodal fusion. Reproduce one mandated baseline (DialogueGCN / MMGCN / GraphCFC), then add at least one required improvement.

Testing

- Metrics: Accuracy, Weighted F1-score, Macro F1-score, and confusion matrices.
- Baseline comparison: compare against the reproduced baseline and at least one additional recent ERC model that uses attention mechanisms or graph learning.
- Ablation and error analysis are mandatory (see Model Evaluation).
- Evaluation protocol: report split, seed, hardware, and training/inference time.

Deployment

- Runtime environment: a GPU-enabled machine (or high-performance CPU) with Python and PyTorch / PyTorch Geometric.
- Interface: a prototype application delivering the enhanced visualizations listed above (timeline, interaction graph, emotion transitions, confidence, attention heatmap).

(*) 3.2. Main proposal content (including result and product)

a. **Theory and practice (document):**

Students should apply their knowledge from AIL302m (Artificial Intelligence), AIP391 (Machine Learning), DPL301m (Deep Learning), NLP-related courses, Computer Vision, and Software Engineering in this project.

Conduct a literature review covering: Emotion Recognition in Conversations, Multimodal Learning, Transformer-based Representation Learning, Graph Neural Networks, Graph Attention Networks, Multimodal Feature Fusion, and Model Evaluation.

Research data-processing and graph-construction algorithms, together with deep-learning models, to improve emotion-recognition accuracy.

- Artificial Intelligence technologies:

Model / framework: PyTorch, PyTorch Geometric (GNN), and HuggingFace Transformers.

Pretrained encoders: RoBERTa/BERT (text), Wav2Vec2/HuBERT (audio), and ViT/CLIP (visual).

Deployment: Python.

b. **Products:**

- A graph-based multimodal emotion recognition model.
- A prototype system for conversational emotion analysis (per-utterance prediction and the enhanced visualizations).
- Experimental evaluation reports, ablation studies, and comparative analysis.
- A complete capstone thesis; results are encouraged to be extended into undergraduate research publications or international conference papers.

Reproducibility package (required):

To make grading transparent and results verifiable, students must provide:

- Source code.
- Trained model weights.
- Configuration files.
- Inference script.
- README with setup and run instructions.
- Experiment log (metrics, seeds, and settings per run).

Proposed Tasks (aligned with the AIP491 15-week schedule):

- **Task package 1 (Weeks 1–3):** Literature review, requirement analysis, dataset exploration, and identification of the research gap and target baseline (→ Report 1: Introduction; Report 3: Existing Systems).
- **Task package 2 (Weeks 4–6):** Data preprocessing, multimodal feature extraction, and reproduction of the chosen baseline (→ Report 4: Methodology).
- **Task package 3 (Weeks 7–10):** Conversation graph construction, GNN implementation, multimodal fusion, and the proposed improvement (→ Report 4/5).

- **Task package 4 (Weeks 11–13):** Model optimization, ablation studies, error analysis, and comparative evaluation (→ Report 6 & 7: Results/Discussion).
- **Task package 5 (Weeks 14–15):** Prototype development, thesis writing, and final presentation (→ Final report and defense).

AI Ethics and Responsible AI:

Because GMERC processes affective (emotional) data, students must include a short discussion of responsible-AI considerations, consistent with AIP491 CLO7 (Professionalism and Ethical Conduct in AI) and the syllabus session on AI Ethics and Data Privacy:

- Privacy of conversational and biometric (voice/face) data, and appropriate handling of benchmark data.
- Emotional bias — the risk that models learn spurious cues (e.g., speaker identity, accent) instead of genuine emotion.
- Fairness across speakers, demographics, and emotion classes (including minority emotions).
- Responsible use and limitations of automated emotion recognition in sensitive contexts.

Expected Learning Outcomes:

Upon completion, students will be able to (mapped to AIP491 CLOs):

- Implement Graph Neural Networks and multimodal AI systems (CLO3, CLO8).
- Design and reproduce a research baseline, then propose and validate a technical improvement (CLO3, CLO8, CLO10).
- Conduct reproducible AI experiments with proper protocol, ablation, and error analysis (CLO2, CLO5, CLO9).
- Critically analyze experimental results and reflect on limitations (CLO9).
- Develop and deploy an interactive AI prototype (CLO3, CLO6).
- Apply professional and ethical standards to affective-computing systems (CLO7).
- Communicate research findings effectively in a written thesis and oral defense (CLO5, CLO6).

References:

Foundations

[1] T. N. Kipf and M. Welling. Semi-Supervised Classification with Graph Convolutional Networks. Proc. ICLR, 2017.

- [2] D. Ghosal, N. Majumder, A. Gelbukh, et al. DialogueGCN: A Graph Convolutional Neural Network for Emotion Recognition in Conversation. Proc. EMNLP-IJCNLP, 2019.
- [3] S. Poria, D. Hazarika, N. Majumder, et al. MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations. Proc. 57th Annual Meeting of the ACL, 2019. (benchmark dataset)

Baseline Models

- [4] J. Hu, Y. Liu, J. Zhao, et al. MMGCN: Multimodal Fusion via Deep Graph Convolution Network for Emotion Recognition in Conversation. Proc. ACL-IJCNLP, 2021.

Recent SOTA

- [5] J. Li, X. Wang, G. Lv, and Z. Zeng. GraphCFC: A Directed Graph Based Cross-Modal Feature Complementation Approach for Multimodal Conversational Emotion Recognition. IEEE Transactions on Multimedia, vol. 26, pp. 77–89, 2024.
- [6] N. Lu, Z. Han, M. Han, and J. Qian. Bi-stream Graph Learning Based Multimodal Fusion for Emotion Recognition in Conversation. Information Fusion, vol. 106, 2024.
- [7] T. Meng, F. Zhang, Y. Shou, et al. Masked Graph Learning with Recurrent Alignment for Multimodal Emotion Recognition in Conversation. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2024.
- [8] H. Zhao, Y. Chen, S. Liu, et al. DAGCCF: Dynamic Alternating Graphs Based Cross-Modal Complementary Fusion for Conversational Emotion Recognition. Neurocomputing, 2025.

4. Other comments (propose all relevant things if any)

.....

....., date / /

Supervisor (If have)

(Sign and full name)

On behalf of the Registers

(Sign and full name)